# STA2002: Probability and Statistics II

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# Part I

# Descriptive Statistics

# Chapter 1

# Central Measures

## 1.1 Introduction

1. summarizing the data with a few figures
2. to see the forest from the trees
3. to see the big picture with the help of a graph or a diagram
4. to allow comparisons between variables

**Single variable data:**

- $n$ individuals
- a variable $Y$ is measured with some units of measurement
- $Y_i$, the value of the variable $Y$ for individual $i$, $1 \leq i \leq n$.
- the index $i$ in $Y_i$ is just for reference. It is a name.

## 1.2 Various means

### 1.2.1 Arithmetic Mean

**Definition:** For the series $Y = \{Y_i\}_{i=1}^n$, the arithmetic mean is

$$\overline{Y} = \frac{Y_1 + Y_2 + \cdots + Y_n}{n} = \frac{1}{n} \sum_{i=1}^{n} Y_i$$

**Example:** $Y = \{-1, 0, 2, 4, 7\}$,

$$\frac{-1+0+2+4+7}{5} = 2.4$$

**Properties of the Arithmetic Mean:**

- preserves the units of measurement.
- it is the unique constant $c$ such that $nc = c+c+\cdots+c = \Sigma_{i=1}^{n} Y_i$.
- assume $Y_1 \leq Y_2 \leq \cdots \leq Y_n$. It is the unique constant $c$ such that there exists an index $j$, such that $Y_j \leq c \leq Y_{j+1}$ and

  $$\Sigma_{i=1}^{j}(c - Y_i) = \Sigma_{i=j+1}^{n}(Y_i - c).$$

  It balances the sums of positive and negative residuals from it.
- preserves linear transformations. For any constants, $a, b$

  $$\overline{a \cdot Y + b} = a \cdot \overline{Y} + b$$

- the mean of the sums equals the sum of the means. For each pair of sequences $\{Y_i\}_{i=1}^{n}, \{X_i\}_{i=1}^{n}$

  $$\overline{X + Y} = \overline{X} + \overline{Y}$$

- sensitive to all entries: if $Y_i$ changes by $\epsilon$, the mean changes by $\epsilon/n$.

- minimizes the sum of the squares of the residuals, namely the penalty function

$$f(x) = \sum_{i=1}^{n} (Y_i - x)^2$$

**Proof.** As $(a-b)^2 = a^2 - 2ab + b^2$,

$$f(x) = \sum_{i=1}^{n} (Y_i - x)^2 = \sum_{i=1}^{n} Y_i^2 - 2x \sum_{i=1}^{n} Y_i + nx^2$$

A "laughing" parabola, $ax^2 + bx + c$, with

$$a = n, \quad b = -2 \sum_{i=1}^{n} Y_i \text{ and } c = \sum_{i=1}^{n} Y_i^2$$

Hence,

$$x_{\min} = -\frac{b}{2a} = \frac{2 \sum_{i=1}^{n} Y_i}{2n} = \frac{\sum_{i=1}^{n} Y_i}{n} = \overline{Y}$$

END

- **Claim:**

$$\overline{Y}^2 \leq \overline{Y^2}$$

The square of the mean is smaller than or equal to the mean of the squares. $\overline{Y}^2 = \overline{Y^2}$ exists if and only if $Y_i = c$, $1 \leq i \leq n$.

**Proof:**

$$0 \leq f(x_{\min}) = \frac{4ac - b^2}{4a} = \frac{4n\sum_{i=1}^{n} Y_i^2 - 4(\sum_{i=1}^{n} Y_i)^2}{4n}$$

$$\frac{n^2\overline{Y^2} - n^2\overline{Y}^2}{n} = n(\overline{Y^2} - \overline{Y}^2).$$

Equality holds if and only if $b^2 = 4ac$ (when the two roots coincide which due to the fact that $f(x) \geq 0$), namely if and only if $4(n\overline{Y})^2 = 4n(n\overline{Y^2})$, as stated. In this case there exists an $x$ such that $f(x) = 0$. But this is possible if and only if all entries in the series have the same value. **End**

### 1.2.2 Geometric Mean

**Definition:** Assume $Y_i \geq 0$, $1 \leq i \leq n$. The geometric mean is

$$\sqrt[n]{Y_1 \cdot Y_2 \cdots Y_n}$$

Define a penalty function:

$$g(x) = \sum_{i=1}^{n} (\log Y_i - \log x)^2$$

**Claim:** The geometric mean minimizes this penalty function.

**Proof.** Denote by $x_{\min}$ the value we are after. Then, by the previous subsection, $\log x_{\min} = \overline{\log Y}$. But

$$\overline{\log Y} = \tfrac{1}{n} (\log Y_1 + \log Y_2 + \cdots + \log Y_n) = \tfrac{1}{n} \log (Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n) =$$

$$= \log \left( (Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n)^{\frac{1}{n}} \right) = \log \sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n}$$

Hence,

$$\log x_{\min} = \overline{\log Y} = \log \sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n}$$

$$\Downarrow$$

$$x_{\min} = \sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n}$$

Note that the base of the logarithm does not matter. **End**

**Properties of the Geometric Mean:**

- preserves the measurement units.
- For a constant $a \geq 0$,

$$\sqrt[n]{aY_1 \cdot aY_2 \cdot \cdots \cdot aY_n} = a\sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n}$$

- But for $b \neq 0$,

$$\sqrt[n]{(Y_1 + b) \cdot (Y_2 + b) \cdot \cdots \cdot (Y_n + b)} \neq \sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n} + b$$

  the geometric mean is **not** linear with respect to addition of a scalar.

- For $\{X_j\}_{j=1}^n$ with $X_j \geq 0$, $1 \leq j \leq n$,

$$\sqrt[n]{X_1Y_1 \cdot X_2Y_2 \cdot \cdots \cdot X_nY_n} = \sqrt[n]{X_1 \cdot X_2 \cdot \cdots \cdot X_n} \cdot \sqrt[n]{Y_1 \cdot Y_2 \cdot \cdots \cdot Y_n}.$$

  Note that the units of measurement here are those to $X$ times those of $Y$.

### 1.2.3 Harmonic Mean

**Definition:** $Y_i > 0$, $1 \leq i \leq n$. The harmonic mean is:

$$\frac{n}{\frac{1}{Y_1} + \frac{1}{Y_2} + \cdots + \frac{1}{Y_n}} = \frac{1}{\overline{\frac{1}{Y}}}$$

in words, this is the reciprocal to the mean of the reciprocals of the entries in the sequence.

Define the penalty function

$$h(x) = \sum_{i=1}^{n} \left( \frac{1}{Y_i} - \frac{1}{x} \right)^2.$$

**Claim:** The harmonic mean minimizes this penalty function $h(x)$.

**Proof:**

The choice $\frac{1}{x} = \overline{\frac{1}{Y}}$ minimizes this function, as explained above regarding the arithmetic mean. Hence,

$$x_{\min} = \frac{1}{\overline{\frac{1}{Y}}} = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \frac{1}{Y_i}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{Y_i}}$$

**End**

**Properties of the harmonic mean, $H(Y)$:**

- preserves the units of measurement.
- for a constant $a$, $H(aY) = aH(Y)$.
- for constants $a$ and $b$, $H(aY + b) \neq aH(Y) + b$
- $H(X + Y) \neq H(X) + H(Y)$.

### 1.2.4 Weighted averages

For the series $w_i \geq 0$, $1 \leq i \leq n$, with $\Sigma_{i=1}^{n} w_i = 1$, referred to as *weights*, we defined the weighted average of the series $Y$ by $\Sigma_{i=1}^{n} w_i Y_i$.

**Examples:**

- uniform weights: $w_i = 1/n$, $1 \leq i \leq n$, $\overline{Y}$, the arithmetic mean
- weights inversely proportional to the entries, $Y > 0$:

$$w_i = \frac{\frac{1}{Y_i}}{\sum_{j=1}^{n} \frac{1}{Y_j}}, \quad 1 \leq i \leq n,$$

the harmonic mean

$$\frac{\sum_{i=1}^{n} \frac{1}{Y_i} Y_i}{\sum_{i=1}^{n} \frac{1}{Y_i}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{Y_i}} \leq \overline{Y}$$

- weights proportional to the entries, $Y \geq 0$:

$$w_i = \frac{Y_i}{\sum_{j=1}^{n} Y_j}, \quad 1 \leq i \leq n$$

leads to the length-bias mean:

$$\frac{\sum_{i=1}^{n} Y_i^2}{\sum_{i=1}^{n} Y_i} = \frac{\overline{Y^2}}{\overline{Y}} \geq \overline{Y}$$

**Example:** Two families, one with two kids, another with three.

What is the average number of kids in a family **per family**?

$$(2+3)/2 = 2.5.$$

What is the average number of kids in a family **per kid**?

$$\frac{2 \times 2 + 3 \times 3}{2+3} = 2.75$$

## 1.3 Choosing which mean to use

**The inequality of the means:**

For $x_i > 0$, $1 \leq i \leq n$,

$$\frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}} \leq \sqrt[n]{x_1 \cdot x_2 \cdot \cdots \cdot x_n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}$$

with equality if and only if all the numbers are equal.

### 1.3.1 Replacing a series with a constant

One more angle to define mean values:

- **Arithmetic mean:** Which constant $c$ satisfies $\sum_{i=1}^{n} X_i = c + c + \cdots + c = nc$?

  **Answer:** the arithmetic mean $c = \overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$.

- **Geometric mean:** Which constant $c$ satisfies $\prod_{i=1}^{n} X_i = c^n = c \times c \times \cdots \times c$?

  **Answer:** the geometric mean $c = \sqrt[n]{\prod_{i=1}^{n} X_i}$ (for elements that are all positive).

- **Harmonic mean:** Which constant $c$ satisfies $\sum_{i=1}^{n} \frac{1}{X_i} = n \times \frac{1}{c} = \frac{1}{c} + \frac{1}{c} + \cdots + \frac{1}{c}$?

  **Answer:** the harmonic mean $c = \frac{n}{\sum_{i=1}^{n} X_i}$.

**Example 1.**

- Yields of a share in three years: 1.1, 1.23 and 0.9. Total yield $1.1 \times 1.23 \times 0.9 = 1.2177$.
- Arithmetic mean yield: $(1.1 + 1.23 + 0.9)/3 = 1.077$.
- A constant yield of 1.077 per year leads to a three-year yield of $1.077^3 = 1.248 > 1.2177$.
- A constant yield of $(1.2177)^{1/3} = 1.068 < 1.077$ leads to a three year-yield of 1.2177

Indeed, as predicted by the inequality of the means, $1.068 < 1.077$.

**Conclusion:** when reporting average yearly yields, use the geometric, not the arithmetic, mean.

**Example 2.**

- A car travels from A to B for 100 kilometers with speed of 100 km/h. It takes 1-hour.
- On the return, the speed is 50 km/h. It takes 2-hours.

**Question:** What is the average speed?

**Answer:** Is it arithmetic mean $(100 + 50)/2 = 75$km/h?
Driving three hours for 75km/h makes the traveled distance 225km, not 200km.

**A Better Answer:** Total distance of 200km covered divided by 3, the total time needed, leads to an average speed of $200/3 = 66.667$km/h. This is the harmonic mean of 100 and 50:

$$\frac{2}{\frac{1}{100} + \frac{1}{50}} = 66.667.$$

**Conclusion:** When reporting the average speed, state the harmonic mean.

Note that arithmetic mean of the speed per hour is 66.667km/h but per kilometer it is 75km/h.

## 1.4 The median

**Definition:** The one in the middle.

- Reorder the series $Y_1 \leq Y_2 \leq \cdots \leq Y_n$
- If $n$ is odd $med(Y) = Y_{(n+1)/2}$
- If $n$ is even, $med(Y)$ is anything between $Y_{n/2}$ and $Y_{n/2+1}$. A popular choice is the arithmetic mean between these two in the middle

**Example:** $Y = \{-1, 0, 2, 4, 7\}$, then $med\,(Y) = 2$.

### 1.4.1 Minimizing a penalty function

$$h\,(x) = |Y_1 - x| + |Y_2 - x| + \cdots + |Y_n - x| = \sum_{i=1}^{n} |Y_i - x| = n \cdot \overline{|Y - x|}.$$

For the above example

$$h\left(x\right)=|-1-x|+|0-x|+|2-x|+|4-x|+|7-x|=\begin{cases}-5x+12 & \text{for } x\leq -1\\ -3x+14 & \text{for } -1\leq x\leq 0\\ -x+14 & \text{for } 0\leq x\leq 2\\ x+10 & \text{for } 2\leq x\leq 4\\ 3x+2 & \text{for } 4\leq x\leq 7\\ 5x-12 & \text{for } 7\leq x\end{cases}$$

The graph for this function is:

- If $n$ is odd the loss function switches the sign of its slope at $Y_{(n+1)/2}$.

  A unique minimum is attained there

- If $n$ is even, the function is flat along the interval $[Y_{n/2}, Y_{n/2+1}]$

  A continuum minima

$$h\left(med(Y)\right) = \sum_{i=1}^{n} |Y_i - med(Y)|$$

- Half of the series is above the median, half is below it
- Half of the absolute values above change sign, the rest do not
- The median disappears in computation:

Assume that $Y_1 \leq Y_2 \leq \cdots \leq Y_n$. Then,

$$h(med(Y)) = \begin{cases} \sum_{i=\frac{n+1}{2}}^{n} Y_i - \sum_{i=1}^{\frac{n-1}{2}} Y_i & \text{if n is odd} \\ \sum_{i=\frac{n}{2}+1}^{n} Y_i - \sum_{i=1}^{\frac{n}{2}} Y_i & \text{if is even} \end{cases}$$

In words, the sum of the upper half minus the sum of the lower half.

**Properties of the Median:**

- preserves the units of measurement
- for constants $a, b$,

$$med\,(aY + b) = a \cdot med\,(Y) + b$$

- $$med\,(X + Y) \neq med\,(X) + med\,(Y)$$

  For example:

$$med\,(1, 2, 4) = 2 \quad \text{and} \quad med\,(1, 3, 2) = 2$$

  but

$$med(1 + 1, 2 + 3, 4 + 2) = med(2, 5, 6) = 5 \neq 2 + 2 = 4$$

- not much sensitivity: As long as you change an entry without crossing the median, the median does not change
- the median is the only choice(s) such that the number of entries above or equal to it, equals the number of entries below or equal to it

**Strategic considerations: An example.**

- $n$ houses are located along a street at positions $x_1 < x_2 < \cdots < x_n$
- a bus stop is to be located somewhere in the middle
- define 'middle' as the **arithmetic mean**
- a problem: the planner does not know the locations
- an option: ask the residents but they may not tell the truth as they wish the bus stop to be located as close as possible to their house
- for example, if all but $i$ tell $y_j$ (true or not), tenant $i$ will report $y_i$ such that $x_i = \frac{\sum_{j=1}^{n} y_j}{n}$, namely $y_i = n \cdot x_i - \sum_{j \neq i}^{n} y_j$
- in particular, if all but one tell the truth, the last one will not do the same
- confused? you should be... The point is that telling the truth is far from being the automatic choice. Individuals may become strategic

- redefine now 'middle' as the **median**
- no matter what others do, it is one's best to report the truth!
- note: reporting other values (data dependent) might also be optimal but the truth is the only choice which works regardless of the data
- all telling the truth is not a goal of its own but it leads to the optimal location when the loss function is $\Sigma_{i=1}^{n}|Y_i - x|$

# Chapter 2

# Dispersion Measures

## 2.1 Introduction

- Two series may share a center but are spread differently: $\{0, 5, 10\}$ is more spread than $\{4, 5, 6\}$
- a central measure needs first to be agreed upon, then define a measure of dispersion around it
- leads to defining a standard measure of being away from the cen-

ter: The less is the dispersion, the more significant is an absolute distance (residual) from the center

**A wrong choice:** The arithmetic mean of the residuals from the arithmetic mean:

$$\overline{Y - \overline{Y}}$$

**It equals zero:**

$$\overline{Y - \overline{Y}} = \frac{\left(Y_1 - \overline{Y}\right) + \left(Y_2 - \overline{Y}\right) + \cdots + \left(Y_n - \overline{Y}\right)}{n} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right) = 0.$$

Or,

$$\overline{Y - \overline{Y}} = \overline{Y} - \overline{Y} = 0.$$

## 2.2 Variance

**Definition:** Make $\overline{Y}$ the center. Then,

$$\mathrm{Var}(Y) = \overline{\left(Y - \overline{Y}\right)^2}.$$

or

$$\mathrm{Var}\,(Y) = \frac{\left(Y_1 - \overline{Y}\right)^2 + \left(Y_2 - \overline{Y}\right)^2 + \cdots + \left(Y_n - \overline{Y}\right)^2}{n} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2$$

**An alternative formula:**

$$\mathrm{Var}(Y) = \overline{Y^2} - \overline{Y}^2$$

**Proof:**

$$\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 = \sum_{i=1}^{n}\left(Y_i^2 - 2Y_i\overline{Y} + \overline{Y}^2\right) = \sum_{i=1}^{n} Y_i^2 - 2\overline{Y}\sum_{i=1}^{n} Y_i + n\overline{Y}^2 =$$

$$= n \cdot \frac{1}{n}\left(\sum_{i=1}^{n} Y_i^2 - 2\overline{Y}\sum_{i=1}^{n} Y_i + n\overline{Y}^2\right) = n\overline{Y^2} - 2n\overline{Y}^2 + n\overline{Y}^2 = n\left(\overline{Y^2} - \overline{Y}^2\right)$$

END

**A known corollary:**

$$\overline{Y^2} \geq \overline{Y}^2$$

with equality if and only if $\mathrm{Var}(Y) = 0$, namely $Y_i = c$, $1 \leq i \leq n$, for some constant $c$.

**Properties of the variance:**

- $\text{Var}(Y) = 0$ if and only if no variability at all. Otherwise, $\text{Var}(Y) > 0$.
- its units of measurement are wrong: the square of the original units.
- for any constant $b$, $\text{Var}(Y + b) = \text{Var}(Y)$

  **Proof:**

- for any constant $\text{Var}(aY) = a^2\text{Var}(Y)$

  **Proof:**

- In summary, for any two constants $a, b$:

$$\text{Var}(aY + b) = a^2\text{Var}(Y).$$

## 2.3 Standard Deviation

**Definition:** The standard deviation of $Y_i$, $1 \leq i \leq n$:

$$\mathrm{SD}(Y) = \sqrt{\mathrm{Var}(Y)}$$

**Properties of the standard deviation:**

- $\mathrm{SD}(Y) = 0$ if and only if $Y_i = c$, $1 \leq i \leq n$. Otherwise, $\mathrm{SD}(Y) > 0$.
- preserves the units of measurement
- For any constant $b$,

$$\mathrm{SD}(Y + b) = \mathrm{SD}(Y)$$

- For any constant $a$,

$$\mathrm{SD}(aY) = |a|\,\mathrm{SD}(Y)$$

Note: $\sqrt{a^2} = |a|$.

- In summary, for any two constants $a$ and $b$,

$$\mathrm{SD}(aY + b) = |a|\,\mathrm{SD}(Y)$$

**Example: A Bernoulli sequence**

**Definition** $I_i \in \{0, 1\}$, $1 \leq i \leq n$.

Used for a 'yes' (then $I_i = 1$) or 'no' (indicator) variables.

Note:

- $\overline{I}$ is the relative frequency of the property in the population
- $I_i^2 = I_i$, $1 \leq i \leq n$

**Claim:** $\mathrm{Var}(I) = \overline{I}(1 - \overline{I})$ and then, $\mathrm{SD}(I) = \sqrt{\overline{I}(1 - \overline{I})}$.

**Proof 1:**

$$\mathrm{Var}(I) = \frac{1}{n}\sum_{i=1}^{n}(I_i - \overline{I})^2 = \frac{1}{n}\left(\sum_{i=1}^{n} I_i^2 - 2\overline{I}\sum_{i=1}^{n} I_i + n\overline{I}^2\right) =$$

$$= \frac{1}{n}\left(\sum_{i=1}^{n} I_i - 2n\overline{I}^2 + n\overline{I}^2\right) = \frac{1}{n}(n\overline{I} - 2n\overline{I}^2 + n\overline{I}^2) = \overline{I} - \overline{I}^2 = \overline{I}(1 - \overline{I}),$$

as required. END

**Proof 2:**

$$\text{Var}(I) = \overline{I^2} - \overline{I}^2 = \overline{I} - \overline{I}^2 = \overline{I}(1 - \overline{I}),$$

as required. END

If you know $\overline{I}$, you can deduce SD($I$). The converse is 'almost' true: given SD($I$), there are two options for $\overline{I}$.

### 2.3.1 Chebyshev Inequality

**Claim:** For a sequence $Y_i$, $1 \leq i \leq n$, with $\overline{Y}$ and SD($Y$), for any $k > 1$, at least a fraction of $1 - \frac{1}{k^2}$ of its entries are at most $\pm k \cdot \text{SD}(Y)$ away from the mean.

**Example:** $k = 3$**.** At least $1 - \frac{1}{9} = \frac{8}{9}$ of the observations are up to 3 standard deviations (up or down) from $\overline{Y}$. Alternatively put, at most $\frac{1}{9}$ of the observations are further away from $\overline{Y}$ by 3SD($Y$) or more.

## 2.4 Standard units and standardization

**Definition:** For $Y_i$, $1 \leq i \leq n$, let

$$Z_i = \frac{Y_i - \overline{Y}}{\text{SD}(Y)},\ 1 \leq i \leq n$$

The linear transformation whch multiplies by $\frac{1}{\text{SD}(Y)}$ and adds $-\frac{\overline{Y}}{\text{SD}(Y)}$, is called **standardization.**

- standard scores are unit free
- the standardization of $Y$ and of $aY + b$ when $a > 0$ is the same

The standard score of an individual says how it does relative to the population the individual belongs to: the sign of $Y_i - \overline{Y}$ says if one is above or below the mean. But it does not say by how much: this value needs to be compare to $\text{SD}(Y)$. A given absolute difference is more

significant when the $\mathrm{SD}(Y)$ is small in comparison with the case where $\mathrm{SD}(Y)$ is.

**Properties of the standard scores:**

1. The mean of the standard scores is 0.

   **Proof:**

   $$\overline{Z} = \frac{\overline{Y}}{\mathrm{SD}(Y)} - \frac{\overline{Y}}{\mathrm{SD}(Y)} = 0.$$

   END

2. $\mathrm{Var}(Z) = \mathrm{SD}(Z) = 1$

   **Proof:**

   $$\mathrm{Var}(Z) = \mathrm{Var}(\tfrac{Y}{\mathrm{SD}(Y)} - \tfrac{\overline{Y}}{\mathrm{SD}(Y)}) = \mathrm{Var}(\tfrac{Y}{\mathrm{SD}(Y)})$$

   $$= \tfrac{1}{\mathrm{SD}^2(Y)}\mathrm{Var}(Y) = \tfrac{\mathrm{Var}(Y)}{\mathrm{Var}(Y)} = 1.$$

   END

**Note:** you can compare the performance of the same individual with respect to two different variables: Do that by comparing the corresponding Z-scores.

## 2.5 Histogram

A histogram is a method for visualizing quantitative data.

**Example:** $Y_i$, $1 \leq i \leq 100$, are the ages of 100 people. We wish to present these data: 22,72,32,.....,41.

**Step 1:** Determine a number of ranges (classes) of values in each of which some observations fall (the ranges can be different in length). Goldilocks rule: not too many classes (as too few, if at all, will be in a class) but not too few (as the presentation will not be informative enough).

**Step 2:** Calculate the relative frequency of each range.

**Step 3:** Create a coordinate system. The $x$ axis in measured by the units of the variable. Mark the ranges as determined above.

**Step 4:** Decide what is the size of the area in the two-dimensional plane you will call one.

**Step 5:** In each range, draw a rectangular with an area which equals its relative frequency.

Note: the height of the rectangular is a by-product to the drawing. For

| Age | Frequency | Relative frequency |
|---|---|---|
| 15-25 | 2 | 0.02 |
| 25-35 | 10 | 0.1 |
| 35-45 | 19 | 0.19 |
| 45-55 | 27 | 0.27 |
| 55-65 | 25 | 0.25 |
| 65-75 | 16 | 0.16 |
| 75-85 | 1 | 0.01 |
| Total | 100 | 1 |

example, consider the height of the rectangle that corresponds to the $45-55$ years old range. Its width is 10 years and its area is 0.27. Thus, its height is $0.27/10 = 0.027$.

Note: when and only when the lengths of the segments are equal, the heights of the rectangles are also proportional to the relative frequency of the group.

**Question:** 0.027 what?

**Answer:** relative frequency per year, called the density. This is the unit of the y-axis.

Note: The height of the rectangle defines the rate of accumulation of the relative frequency per unit of $x$. The density does not equal the relative frequency. For example, a rectangular height of 0.019 means that each time we move a unit along the $x$-axis in the corresponding range, we accumulate another 1.9% of relative frequency. In other words, the density can be looked at as the derivative of the cumulative relative frequency function.

**Polygon**

Drawing a polygon:

- add two classes, one at the left edge, another at right age
- their length will equal that of the adjacent range
- give them height of zero
- connect the center points of the upper face of the rectangle

Note: The size of the area under the polygon is preserved to one.

# Chapter 3

# Relations between two variables and the regression line

## 3.1 Introduction

Two series of variables $(X_i, Y_i)$, $1 \leq i \leq n$. The order is not important but the pair is. Each variable can be summarized separately.

Possible (vague) questions:

- how the variables are related?
- can we learn from one on the (approximate) value of the other?
- can we say that if one goes up, the same is the case with the other? Or maybe the opposite is the case? or maybe we can say nothing?
- knowing $X_{n+1}$, can we predict $Y_{n+1}$?

## 3.2 Covariance

**Definition:** For $(X_i, Y_i)$, $1 \leq i \leq n$, define the covariance: the mean of the product of the residuals from the mean

$$\mathrm{Cov}(X,Y) = \frac{1}{n}\left[\left(X_1 - \overline{X}\right)\left(Y_1 - \overline{Y}\right) + \cdots + \left(X_n - \overline{X}\right)\left(Y_n - \overline{Y}\right)\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)$$

$$= \overline{\left(X - \overline{X}\right)\left(Y - \overline{Y}\right)}$$

**Example:**

| $i$ | $X_i$ | $Y_i$ | $X_i - \overline{X}$ | $Y_i - \overline{Y}$ | $\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)$ |
|---|---|---|---|---|---|
| 1 | 1 | 5 | $-3$ | $-1$ | 3 |
| 2 | 3 | 4 | $-1$ | $-2$ | 2 |
| 3 | 4 | 7 | 0 | 1 | 0 |
| 4 | 5 | 1 | 1 | $-5$ | $-5$ |
| 5 | 7 | 13 | 3 | 7 | 21 |

$\overline{X} = 4, \overline{Y} = 6, \mathrm{SD}(X) = 2, \mathrm{SD}(Y) = 4$

$\mathrm{Cov}(X, Y) = 4.2 > 0.$

In general, when one variable goes up, the same is the case with the other but this is not a precise statement: see $i = 4$.

**Formula:**

$$\mathrm{Cov}(X,Y) = \overline{X \cdot Y} - \overline{X} \cdot \overline{Y}$$

**Proof:**

$$\mathrm{Cov}(X,Y) = \tfrac{1}{n} \textstyle\sum_{i=1}^{n} \left(X_i - \overline{X}\right) \left(Y_i - \overline{Y}\right)$$

$$= \frac{1}{n} \sum_{i=1}^{n} X_i Y_i - \frac{1}{n} \sum_{i=1}^{n} X_i \overline{Y} - \frac{1}{n} \sum_{i=1}^{n} \overline{X} Y_i + \frac{1}{n} \sum_{i=1}^{n} \overline{X} \cdot \overline{Y}$$

$$= \tfrac{1}{n} \textstyle\sum_{i=1}^{n} X_i \cdot Y_i - \overline{Y} \cdot \overline{X} - \overline{X} \cdot \overline{Y} + \overline{X} \cdot \overline{Y} = \overline{X \cdot Y} - \overline{X} \cdot \overline{Y}$$

END

**Definition:**

$X$ and $Y$ are positively (negatively) correlated if $\mathrm{Cov}(X,Y)$ is positive (negative). They are uncorrelated if $\mathrm{Cov}(X,Y) = 0$.

**Properties of the covariance:**

1. the covariance is measured by the product of the individual units of measurement. For example, if $X$ is in meters and $Y$ in kg, then $\mathrm{Cov}(X,Y)$ is in meters times kg.

2. for any constants $a, b$,

$$\mathrm{Cov}(X + a, Y + b) = \mathrm{Cov}(X, Y)$$

3. for any constants $a, b$,

$$\mathrm{Cov}(aX, bY) = ab \cdot \mathrm{Cov}(X, Y)$$

4. in summary, for any constants $a, b, c, d$,

$$\mathrm{Cov}(aX + b, cY + d) = ac \cdot \mathrm{Cov}(X, Y).$$

5.

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

6.

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\mathrm{Cov}(X, Y)$$

**Proof:**

$$\mathrm{Var}(X \pm Y) = \frac{1}{n}\sum_{i=1}^{n}\left[(X_i \pm Y_i) - \left(\overline{X \pm Y}\right)\right]^2$$

$$= \tfrac{1}{n}\textstyle\sum_{i=1}^{n}\left[\left(X_i - \overline{X}\right) \pm \left(Y_i - \overline{Y}\right)\right]^2$$

$$= \tfrac{1}{n}\textstyle\sum_{i=1}^{n}\left[\left(X_i - \overline{X}\right)^2 + \left(Y_i - \overline{Y}\right)^2 \pm 2\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)\right]$$

$$= \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\mathrm{Cov}(X, Y).$$

END

7.

$$\mathrm{Cov}(X + Y, Z) = \mathrm{Cov}(X, Z) + \mathrm{Cov}(Y, Z)$$

## 3.3 Correlation Coefficient

**Definition:** The correlation coefficient between $X$ and $Y$:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X) \cdot \mathrm{SD}(Y)}$$

**Remark:**

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\text{SD}(X)\cdot\text{SD}(Y)} = \frac{1}{n}\sum_{i=1}^{n}\frac{X_i-\overline{X}}{\text{SD}(X)}\cdot\frac{Y_i-\overline{Y}}{\text{SD}(Y)}$$

Note: This is the covariance between the standard scores of $X$ and the standard scores of $Y$.

**Properties of the Correlation Coefficient:**

1. It is free of units of measurement

2. $$\text{Corr}(X,Y) = \text{Corr}(Y,X)$$

3. For any constants $a, b, c, d$,

$$\text{Corr}(aX+b, cY+d) = \frac{ac\text{Cov}(X,Y)}{|a|\,\text{SD}(X)\,|c|\,\text{SD}(Y)} = \begin{cases} \text{Corr}(X,Y) & \text{if } a\cdot c > 0 \\ -\text{Corr}(X,Y) & \text{if } a\cdot c < 0 \end{cases}$$

and in particular:

$$\text{Corr}\left(\frac{X-\overline{X}}{\text{SD}(X)}, \frac{Y-\overline{Y}}{\text{SD}(Y)}\right) = \text{Corr}(X,Y).$$

The following is a fundamental property of the correlation coefficient:

$$-1 \leq \mathrm{Corr}(X, Y) \leq 1$$

**Proof.** For any scalar $a$,

$$0 \leq \mathrm{Var}(aX - Y) = a^2\mathrm{Var}(X) - 2a\mathrm{Cov}(X, Y) + \mathrm{Var}(Y).$$

Since we have a quadratic function without a root, we conclude

$$\mathrm{Cov}^2(X, Y) \leq \mathrm{Var}(X)\mathrm{Var}(Y),$$

which is equivalent to what we need to prove. **End**

Equality exists (and then there is a single root) if and only if there exists an $a$ with $\mathrm{Var}(aX - Y) = 0$, namely for this value for $a$, $Y_i - aX_i$, $1 \leq i \leq n$ is a constant. Call this constant $b$. Then, $Y_i = aX_i + b$, $1 \leq i \leq n$, namely all the $n$ points are placed on some line. Moreover, this is

the case if and only if $\text{Corr}(X, Y) = 1$ or $-1$, namely $\text{Corr}^2(X, Y) = 1$. Moreover, the sign of the slope $a$ coincides then with the sign of the correlation.

It follows that when the correlation coefficient is e.g. 0.8, there is a strong relationship between the two variables.

The sign of the correlation coefficient can sometimes be guessed by inspecting the scatter diagram.

## 3.4 Linear Regression

### 3.4.1 Linear functions

Each one of the following defines a line:

- two points it crosses
- a point it crosses and its slope
- its slope, $a$, and its intersection (of the y-axis) $b$: $y = ax + b$

Unit of measurements:

- $b$ shares units with the $y$-variable,
- $a$ comes with the ratio between those of $y$ and those of $x$

### 3.4.2 The least squares line

The two-variable series $(X_i, Y_i)$, $1 \leq i \leq n$, are given.

Usually, they are not sitting on a single line.

**Question:** Which line represents these data best?

**Answer:** There is more than one answer...(as in the case of central measures)

One option:

$$(a^*, b^*) = \arg\min_{a,b} \sum_{i=1}^{n} (Y_i - (b + aX_i))^2$$

Minimizes the sum of the squares of the vertical residuals.

**Note:** no symmetry between $X$ and $Y$. We wish to learn from $X$ on $Y$ (but we can learn form $Y$ on $X$, if we reverse the roles)

**Note:** A possible, sub-optimal, choice is $a = 0$ and $b = \overline{Y}$. It is the best choice among all lines with a zero slope (those which are parallel to the x-axis). The penalty equals

$$\sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2 = n \cdot \mathrm{Var}(Y).$$

But can we do better?

**Terminology:** The least square line is also called the **regression** line of $Y$ on $X$.

Why regression?? An answer will be given later.

**Claim:** The regression line of $Y$ on $X$ is the unique line which passes through $\left(\overline{X}, \overline{Y}\right)$ with a slope of $a = \text{Corr}(X,Y) \cdot \frac{\text{SD}_{(Y)}}{\text{SD}_{(X)}}$, or equivalently: the unique one whose intercept is $b = \overline{Y} - a\overline{X}$ and whose slope is:

$$a = \text{Corr}(X,Y) \cdot \frac{\text{SD}(Y)}{\text{SD}(X)}$$

**Note:**

$$a = \text{Corr}(X,Y) \cdot \frac{\text{SD}(Y)}{\text{SD}(X)} = \frac{\text{Cov}(X,Y)}{\text{SD}(X)\text{SD}(Y)} \cdot \frac{\text{SD}(Y)}{\text{SD}(X)} = \frac{\text{Cov}(X,Y)}{\text{Var}(X)}.$$

**Proof 1:** For any given slope $a$,

$$\overline{Y} - a\overline{X} = \arg\min_b \sum_{i=1}^{n} (Y_i - b - aX_i)^2$$

From now on, we replace $b$ with this value, making the optimization problem that of a one variable $a$:

$$\sum_{i=1}^{n} \left(Y_i - \left(\overline{Y} - a\overline{X}\right) - aX_i\right)^2 = \sum_{i=1}^{n} \left[\left(Y_i - \overline{Y}\right) - a\left(X_i - \overline{X}\right)\right]^2$$

$$= \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2 - 2a \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right) + a^2 \sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2$$

a laughing parabola when the variable is $a$.

$$a_{\min} = \frac{2\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right)}{2\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}$$

$$= \frac{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right)}{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$$

**END**

**Proof 2.** First note that

$$\overline{(Y - b - aX)^2} = \mathrm{Var}(Y - b - aX) + \overline{Y - b - aX}^2.$$

For any choice for $a$, the choice for $b$ such that $b = \overline{Y} - a\overline{X}$ zeros the second term, making it the lowest possible value. The first term is not a function of $b$, so all left is to minimize $\mathrm{Var}(Y - ax)$ with respect to $a$. Indeed, $\mathrm{Var}(Y - aX) = \mathrm{Var}(Y) - 2a\mathrm{Cov}(X,Y) + a^2\mathrm{Var}(X)$, which is a quadratic function of in $a$. In particular, it is minimized at

$$a_{min} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)},$$

as required.

**Claim:** For the optimal choice for $a$ and $b$, let $\hat{Y}_i = aX_i + b$, $1 \leq i \leq n$,

called the **fitted** value. Then,

$$\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = n\text{Var}(Y)[1 - \text{Corr}^2(X,Y)].$$

**Proof.** Note for $y = cx^2 + dx + e$, $c > 0$ (laughing parabola), $y_{\min} = e - \frac{d^2}{4c}$. For our case,

$$\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 - \frac{\left[-2\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right)\right]^2}{4\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}$$

$$= \sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 - \frac{\left[\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right)\right]^2}{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}$$

$$= \sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 \left[1 - \frac{\left[\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)\left(X_i - \overline{X}\right)\right]^2}{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 \cdot \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}\right]$$

$$= \sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2 \left[1 - \text{Corr}^2(X,Y)\right] = n\text{Var}(Y)[1 - \text{Corr}^2(X,Y)]$$

**End**

**Remark:** Alternative proof for the fact that $\text{Corr}^2(X,Y) \leq 1$.

**Remark:** $0 = \sum_{i=1}^{n}(Y_i - b - aX_i)^2$, namely all points are located on a line, if and only if $\text{Corr}^2(X,Y) = 1$.

**Note:**

$$\frac{\sum_{i=1}^{n}(Y_i-\hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i-\overline{Y})^2}=\frac{n\mathrm{Var}(Y)(1-\mathrm{Corr}^2(X,Y))}{n\mathrm{Var}(Y)}=1-\mathrm{Corr}^2(X,Y).$$

The lower is this value, the better $X$ explains the variability of $Y$ with the use of a linear function.

**Notations:** The abbreviation $SS$ is used in statistics for Sum of Squares.

$$S_{xx}=\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2=n\mathrm{Var}(X)$$

$$S_{yy}=\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2=n\mathrm{Var}(Y)$$

$$S_{xy}=\sum_{i=1}^{n}\left(X_i-\overline{X}\right)\left(Y_i-\overline{Y}\right)=n\mathrm{Cov}(X,Y)$$

$$\hat{Y}_i=b+aX_i, 1\leq i\leq n, \qquad \hat{y}=b+ax \quad , \quad S_{ee}=\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i-\overline{Y-\hat{Y}}\right)^2=\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2$$

$$S_{\hat{y}\hat{y}}=\sum_{i=1}^{n}\left(\hat{Y}_i-\overline{Y}\right)^2$$

Note that with these notations, the slope of the regression line is

$$a = \frac{S_{xy}}{S_{xx}}.$$

Let us consider the degree of proximity of the points to the regression line compared to the degree of proximity of the points to the constant line of the mean of $Y$: Denote

$$\mathrm{Corr}^2(X, Y) = R^2$$

and note that with those notations it holds that:

$$1 - R^2 = \frac{S_{ee}}{S_{yy}}$$

**Properties of the Regression Line:**

1. The units of measurement of the intercept are as of $Y$ and that of $a$ are as the ratio between the units of $Y$ and those of $X$.
2. The line passes through the point $\left(\overline{X}, \overline{Y}\right)$.

3. For the residuals $e_i = Y_i - \hat{Y}_i$, $1 \leq i \leq n$,

$$\sum_{i=1}^{n} e_i = 0 \quad \text{or equivalently,} \quad \overline{e} = 0.$$

**Proof:**

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right) = \sum_{i=1}^{n} (Y_i - b - aX_i) = \sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} b - \sum_{i=1}^{n} aX_i =$$

$$= n\overline{Y} - nb - an\overline{X} = n\overline{Y} - n\left(\overline{Y} - a\overline{X}\right) - an\overline{X} = 0.$$

**End**

4. The mean of the fitted values equals to the mean of the real values:

$$\overline{\hat{Y}} = \overline{Y}$$

**Proof:**

$$\overline{\hat{Y}} = \overline{aX + b} = a\overline{X} + b = \overline{Y}$$

**End**

**Remark:** Properties $1 - 4$ hold for any line passing through the point $(\overline{X}, \overline{Y})$, and not only for the regression line.

5.

$$\sum_{i=1}^{n} X_i e_i = 0, \text{ or equivalently, } \overline{Xe} = 0.$$

The weighted average of the residuals, with weights proportional to the X value (which can be negative) is zero.

**Proof:**

$$\sum_{i=1}^{n} X_i e_i = \sum_{i=1}^{n} X_i \left(Y_i - \hat{Y}_i\right) = \sum_{i=1}^{n} X_i \left(Y_i - b - aX_i\right)$$

$$= \textstyle\sum_{i=1}^{n} X_i \left(Y_i - \overline{Y} + a\overline{X} - aX_i\right) = \sum_{i=1}^{n} X_i \left(Y_i - \overline{Y}\right) - a \sum_{i=1}^{n} X_i \left(X_i - \overline{X}\right)$$

$$= \textstyle\sum_{i=1}^{n} \left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right) - a \sum_{i=1}^{n} \left(X_i - \overline{X}\right)\left(X_i - \overline{X}\right)$$

$$= \textstyle\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y}) - a \sum_{i=1}^{n} (X_i - \overline{X})^2$$

$$= n\mathrm{Cov}(X,Y) - \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} n\mathrm{Var}(X) = n\mathrm{Cov}(X,Y) - n\mathrm{Cov}(X,Y) = 0$$

**End**

6.

$$\sum_{i=1}^{n} \hat{Y}_i e_i = 0, \quad \text{or equivalently, } \overline{\hat{Y}e} = 0$$

**Proof:**

$$\sum_{i=1}^{n} \hat{Y}_i e_i = \sum_{i=1}^{n} (b + aX_i)\, e_i = b \sum_{i=1}^{n} e_i + a \sum_{i=1}^{n} X_i e_i = b \cdot 0 + a \cdot 0 = 0$$

**Note:** As in point 4, but with weights $\hat{Y}_i, 1 \leq i \leq n$.

7.

$$\sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2 = \sum_{i=1}^{n} \left(\hat{Y}_i - \overline{Y}\right)^2 + \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

or

$$S_{yy} = S_{\hat{y}\hat{y}} + S_{ee}$$

The total variability equals the explained variability plus the unexplained variability.

**Proof:**

$$\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2=\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i+\hat{Y}_i-\overline{Y}\right)^2$$

$$=\sum_{i=1}^{n}\left[\left(Y_i-\hat{Y}_i\right)^2+2\left(Y_i-\hat{Y}_i\right)\left(\hat{Y}_i-\overline{Y}\right)+\left(\hat{Y}_i-\overline{Y}\right)^2\right]$$

$$=\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2+2\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)\left(\hat{Y}_i-\overline{Y}\right)+\sum_{i=1}^{n}\left(\hat{Y}_i-\overline{Y}\right)^2$$

$$=\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2+\sum_{i=1}^{n}\left(\hat{Y}_i-\overline{Y}\right)^2$$

since

$$\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)\left(\hat{Y}_i-\overline{Y}\right)=\sum_{i=1}^{n}e_i\hat{Y}_i-\sum_{i=1}^{n}e_i\overline{Y}=0-\overline{Y}\sum_{i=1}^{n}e_i=0$$

**End**

**Note:**

(a) the total variability is not a function of $X$

(b) the higher the explained variability is, namely the higher $\text{Corr}^2(X,Y)$ is, the better

8. $y = ax + b$ is as

$$\frac{y - \overline{Y}}{\mathrm{SD}(Y)} = \mathrm{Corr}(X, Y) \cdot \frac{x - \overline{X}}{\mathrm{SD}(X)}$$

or

$$Z_Y = \mathrm{Corr}(X, Y) Z_X$$

This is the regression line of the standardization of $Y$ on the standardization of $X$.

9. $\mathrm{Corr}^2(X, Y)$ measures how dense the points around the regression line are, compared to their density around the (constant) line $y = \overline{Y}$.

10.

$$\mathrm{Var}(\hat{Y}) = \mathrm{Corr}^2(X, Y)\mathrm{Var}(Y)$$

**Proof.** First, observe that

$$\mathrm{Var}(\hat{Y}) = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - \bar{\hat{Y}})^2 = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2.$$

The rest is now immediate from point 7 above and the already proved fact that $\Sigma_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = (1 - \mathrm{Corr}^2(X, Y))\Sigma_{i=1}^{n}(Y_i - \bar{Y})^2$.

**End**

**Conclusion:** The larger the $\text{Corr}^2(X,Y)$ is, the lower the dispersion of points around the regression line, compared to their scatter around the average line (scatter in the sense of the sum of the squares of deviations). To sum up: The larger the correlation is, the regression line of $Y$ on $X$ is a better predictor for $Y$ in comparison to the mean, $\overline{Y}$ (which ignores the value of $X$)..

## 3.5 Regression to the Mean

The regression line

$$\frac{y-\overline{Y}}{\text{SD}(Y)} = \text{Corr}(X,Y)\frac{x-\overline{X}}{\text{SD}(X)}$$

Let $x = \overline{X} + k\text{SD}(X)$ (namely, $Z_x = k$), then the fitted value equals $\hat{y} = \overline{Y} + k\text{Corr}(X,Y)\text{SD}(Y)$, (namely, $Z_{\hat{y}} = k\text{Corr}(X,Y)$).

For a father who is $k$ standard deviations from his population's mean, his daughter is $\text{Corr}(X,Y)k$ standard deviations away from her population's mean.

**The line of standard deviations:** Define the line of standard

deviations to be $\frac{y-\overline{Y}}{\text{SD}(Y)} = \frac{x-\overline{X}}{\text{SD}(X)}$. Equivalently:

$$y = \frac{\text{SD}(Y)}{\text{SD}(X)}(x - \overline{X}) + \overline{Y} = \frac{\text{SD}(Y)}{\text{SD}(X)}x - \frac{\text{SD}(Y)}{\text{SD}(X)}\overline{X} + \overline{Y}$$

The line of standard deviations is not characterized by the regression to the mean and it defines a full correlation. That is, in the prediction determined by the line of standard deviations, if $X_i$ is $k$ standard deviations above (below) $\overline{X}$, then $\hat{Y}$ is also $k$ standard deviations above (below) $\overline{Y}$.

Note that the regression line is the one that minimizes the residuals so that the line of standard deviations leads to an inferior, i.e., larger, $S_{ee}$.

**The regression line of $X$ on $Y$**

1. The regression line of $Y$ on $X$:

$$\frac{y-\overline{Y}}{\mathrm{SD}(Y)} = \mathrm{Corr}(X,Y)\frac{x-\overline{X}}{\mathrm{SD}(X)}.$$

   The regression line of $X$ on $Y$:

$$\frac{x-\overline{X}}{\mathrm{SD}(X)} = \mathrm{Corr}(X,Y)\frac{y-\overline{Y}}{\mathrm{SD}(Y)}$$

   In the same coordinate system:

$$\frac{y-\overline{Y}}{\mathrm{SD}(Y)} = \frac{1}{\mathrm{Corr}(X,Y)}\frac{x-\overline{X}}{\mathrm{SD}(X)}$$

2. The two lines meet at $(\overline{X},\overline{Y})$.

   The slope of the regression line of $X$ on $Y$, $\frac{\mathrm{SD}(Y)}{\mathrm{SD}(X)}\frac{1}{\mathrm{Corr}(X,Y)}$, is sharper than the slope of the regression line of $Y$ on $X$, $\mathrm{Corr}(X,Y)\frac{\mathrm{SD}(Y)}{\mathrm{SD}(X)}$.

3. The slope of the line of the standard deviations, $\mathrm{SD}(Y)/\mathrm{SD}(X)$, is somewhere in between.

4. Recall

$$\frac{S_{ee}}{S_{yy}} = 1 - \mathrm{Corr}^2(X,Y)$$

Define $\hat{X}_i$, $1 \leq i \leq n$, the fitted value of the regression line of $X$ on $Y$.

Define $S^*_{ee} = \sum_{i=1}^{n}(X_i - \hat{X}_i)^2$.

Then,

$$\frac{S^*_{ee}}{S_{xx}} = 1 - \text{Corr}^2(X, Y)$$

and

$$\frac{S^*_{ee}}{S_{ee}} = \frac{S_{xx}}{S_{yy}} = \frac{\text{Var}(X)}{\text{Var}(Y)}.$$

5. The ratios of the explained variances from the totals, are the them. The prediction values of one variable on the other, are the same.

**Remark: Correlation is not causality.**

1. Had it been, we would land in a circular argument

2. There might be a third factor which explains the two

# Part II

# A review: Random variables and distributions

# Chapter 4

# Joint, marginal and conditional distributions

## 4.1 Joint and marginal distributions: An example

Consider the following joint distribution between two random variables $X$ and $Y$:

| X \ Y | 1 | 2 | 3 | |
|---|---|---|---|---|
| 1 | 0 | 1/6 | 1/6 | 1/3 |
| 2 | 1/6 | 0 | 1/6 | 1/3 |
| 3 | 1/6 | 1/6 | 0 | 1/3 |
| | 1/3 | 1/3 | 1/3 | 1 |

You can imagine putting three balls in a bag, where the number of dots appearing on them are one, two and three. One draws two balls without replacement. Then, $X$ denotes the number of dots appearing on the first drawn ball and $Y$ states the number on the second.

The entries in the square define the *joint distribution* $\mathrm{P}(X = 1, Y = 1) = 0$. On both the horizontal and vertical margins we have the *marginal distributions*. For example,

$$\mathrm{P}(X = 2) = \mathrm{P}(X = 2, Y = 1) + \mathrm{P}(X = 2, Y = 2) + \mathrm{P}(X = 2, Y = 3) = \frac{1}{6} + 0 + \frac{1}{6} = \frac{1}{3}.$$

Note that the primitive of this model are the joint probabilities, from which all other parameters can be derived.

Observe that $X$ and $Y$ are *identically distributed*. This means that they both follow the same probability law. It says nothing on being identical. In fact, in this example $\mathrm{P}(X = Y) = 0$.

The computation of the expected value and of the variance is immediate:

$$\mathrm{E}(X) = \sum_{i=1}^{3} i\mathrm{P}(X=i) = 1\frac{1}{3} + 2\frac{1}{3} + 3\frac{1}{3} = 2$$

and

$$\mathrm{Var}(X) = \sum_{i=1}^{3} (i-2)^2\mathrm{P}(X=i) = 1^2\frac{1}{3} + 0^2\frac{1}{3} + 1^2\frac{1}{3} = \frac{2}{3}.$$

Finally, $\mathrm{SD}(X) = \sqrt{\mathrm{Var}X} = \sqrt{\frac{2}{3}}$. Note that as $X$ and $Y$ are identically distributed, the random variable $Y$ comes with the same parameters: $\mathrm{E}(Y) = 2$, $\mathrm{Var}(Y) = 2/3$ and $\mathrm{SD}(Y) = \sqrt{\frac{2}{3}}$. Note the expected value and the standard deviation preserves the unit of measurement of the random variable, while the variance is in the square of these units.

Consider now the random variable $X + Y$. You can see that it gets the values of 3,4 and 5 with probabilities $1/3$ each. Hence $\mathrm{E}(X+Y) = 4$. As we can see $\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$. But this is not coincidental: It holds in general.

**Proof.**

$$\mathrm{E}(X+Y) = \sum_y \sum_x (x+y)\mathrm{P}(X=x, Y=y) = \sum_y \sum_x x\mathrm{P}(X=x, Y=y) + \sum_y \sum_x y\mathrm{P}(X=x, Y=y)$$

$$= \sum_x x \sum_y \mathrm{P}(X=x, Y=y) + \sum_y y \sum_x \mathrm{P}(X=x, Y=y) = \sum_x x\mathrm{P}(X=x) + \sum_y y\mathrm{P}(Y=y) = \mathrm{E}(X) + \mathrm{E}(Y).$$

As for $\mathrm{Var}(X+Y)$, note that its distribution is as follows:

| $X+Y$ | 3 | 4 | 5 |
|---|---|---|---|
| $\mathrm{P}(X+Y)$ | 1/3 | 1/3 | 1/3 |

Note that $X+Y$ and $X+2$ are identically distributed. Hence, $\mathrm{Var}(X+Y) = \mathrm{Var}(X+2) = \mathrm{Var}(X) = 2/3$. Finally, define $Z$ the value of the left ball. Clearly, $Z$ is distributed as $X$ (and as $Y$). Alternatively, $Z = 6-(X+Y)$. Hence, $\mathrm{E}(Z) = \mathrm{E}(6-X-Y) = 6-\mathrm{E}(X)-\mathrm{E}(Y) = 6-2-2 = 2$. Likewise, $\mathrm{Var}(Z) = \mathrm{Var}(6-(X+Y)) = \mathrm{Var}(-1(X+Y)) = \mathrm{Var}(X+Y) = 2/3$.

Recall that $\arg\min_a \mathrm{E}(X-a)^2 = \mathrm{E}(X)$ and hence $\min_a \mathrm{E}(X-a)^2 = \mathrm{E}(X-\mathrm{E}(X))^2 = \mathrm{Var}(X)$.

## 4.2 Conditional distributions

Using Bayes' rule we can see that

$$\mathrm{P}(X=1|Y=2) = \frac{\mathrm{P}(X=1,Y=2)}{\mathrm{P}(Y=2)} = \frac{1/6}{1/3} = \frac{1}{2}.$$

This is the *conditional probability* that $X=1$ given $Y=2$. We can repeat that for all possible values for $X$ and get the *conditional distri-*

*bution* of $X$ given $Y = 2$:

$$\mathrm{P}(X = 1|Y = 2) = \frac{1}{2},\ \mathrm{P}(X = 2|Y = 2) = 0 \text{ and } \mathrm{P}(X = 3|Y = 2) = \frac{1}{2}.$$

As for any distribution, we can define the *conditional expectation* of $X$ given $Y = 2$:

$$\mathrm{E}(X|Y = 2) = 1\frac{1}{2} + 2 \times 0 + 3\frac{1}{2} = 2.$$

Note that this conditional expected value coincides with the unconditional one but this does not mean that this is always the case. The conditional distribution for $X$ given $Y = 1$ and the conditional distribution of $X$ given $Y = 3$ can de derived in a similar way. We leave the details to you. It is then possible to see that

$$\mathrm{E}(X|Y = 1) = 2.5,\ \ \mathrm{E}(X|Y = 2) = 2 \ \text{ and } \mathrm{E}(X|Y = 3) = 1.5.$$

You can consider the conditional means as a way to update one's rough idea on $X$ from $\mathrm{E}(X)$ to the corresponding conditional mean, once the value of $Y$ is revealed.

Since $Y$ is a random variable, we can conclude that $\mathrm{E}(X|Y)$, which is itself a random variable as any other function of $Y$, has its own

distribution. Here the possible values for $\mathrm{E}(X|Y)$ are 1.5, 2 and 2.5 and their corresponding probabilities are 1/3, 1/3 and a 1/3, respectively. We can hence pose the following question: What is the expected value of the conditional expected value of $\mathrm{E}(X|Y)$. The answer is

$$\mathrm{E}(\mathrm{E}(X|Y)) = \sum_y \mathrm{E}(X|Y=y)\mathrm{P}(Y=y) = 1.5\frac{1}{3} + 2\frac{1}{3} + 2.5\frac{1}{3} = 2.$$

This value also coincides with $\mathrm{E}(X)$ but this is not a coincidence: We know that for any pair of random variables $X$ and $Y$:

$$\mathrm{E}(\mathrm{E}(X|Y)) = \mathrm{E}(X). \tag{4.1}$$

This is the *law of total expectation* or the *law of iterated expectations*.

*Proof.*

$$\mathrm{E}(\mathrm{E}(X|Y)) = \sum_y \mathrm{E}(X|Y=y)\mathrm{P}(Y=y) = \sum_y \sum_x x\mathrm{P}(X=x|Y=y)\mathrm{P}(Y=y)$$

$$= \sum_y \sum_x x\frac{\mathrm{P}(X=x,Y=y)}{\mathrm{P}(Y=y)}\mathrm{P}(Y=y) = \sum_y \sum_x x\mathrm{P}(X=x,Y=y)$$

$$= \sum_x x \sum_y \mathrm{P}(X=x,Y=y) = \sum_x x\mathrm{P}(X=x) = \mathrm{E}(X).$$

□

**Remark.** Re-examine (4.1). Was this result expected? Suppose for some pair of random variables $X$ and $Y$, $\mathrm{E}(\mathrm{E}(X|Y)) > \mathrm{E}(X)$. This would say that the expected value of our prediction on $X$ is higher than the expected value of $X$. In other words, just because some other random variable $Y$ exists, we update highly our view on $X$. This does not make sense. True, an *actual realization* of $Y$ can update our view on $X$ to be higher or lower than $\mathrm{E}(X)$, but not the fact that such a random variable $Y$ exists.

By the same token, we can compute the conditional variances. Indeed,

$$\mathrm{Var}(X|Y=1) = (1-2.5)^2 0 + (2-2.5)^2 \frac{1}{2} + (3-2.5)^2 \frac{1}{2} = \frac{1}{4},$$

$$\mathrm{Var}(X|Y=2) = (1-2)^2 \frac{1}{2} + (2-2)^2 0 + (3-2)^2 \frac{1}{2} = 1,$$

and

$$\mathrm{Var}(X|Y=3) = \mathrm{Var}(X|Y=1) = \frac{1}{4}.$$

Note that the conditional variance may sometimes get values larger then the unconditional one and sometimes smaller values. The conditional

variance is also a random variable and its mean can also be computed:

$$\mathrm{E}(\mathrm{Var}(X|Y)) = \sum_y \mathrm{Var}(X|Y=y)\mathrm{P}(Y=y) = \frac{1}{4}\cdot\frac{1}{3} + 1\cdot\frac{1}{3} + \frac{1}{4}\cdot\frac{1}{3} = \frac{1}{2}.$$

This value is smaller than $\mathrm{Var}(Y)$ and this is always the case as we argue shortly. The value of $\mathrm{E}(\mathrm{Var}(X|Y))$ is known as the *variance within* as it summarizes the individual variances for different values of $Y$.

We can also define the *variance between*:

$$\mathrm{Var}(\mathrm{E}(X|Y)) = \sum_y (\mathrm{E}(X|Y=y) - \mathrm{E}(\mathrm{E}(X|Y)))^2\mathrm{P}(Y=y) = \sum_y (\mathrm{E}(X|Y=y) - \mathrm{E}(X))^2\mathrm{P}(Y=y).$$

This value summarizes the dispersion of the value of $\mathrm{E}(X|Y)$ across values of $Y$ (which can be looked at as *strata*). Back to the example:

$$\mathrm{Var}(\mathrm{E}(X|Y)) = (1.5-2)^2\frac{1}{3} + (2-2)^2\frac{1}{3} + (2.5-2)^2\frac{1}{3} = \frac{1}{6}.$$

Observe that

$$\mathrm{Var}(\mathrm{E}(X|Y)) + \mathrm{E}(\mathrm{Var}(X|Y)) = \frac{1}{6} + \frac{1}{2} = \frac{2}{3} = \mathrm{Var}(X).$$

This is not coincidental. The following is well-known:

$$\mathrm{Var}(X) = \mathrm{Var}(\mathrm{E}(X|Y)) + \mathrm{E}(\mathrm{Var}(X|Y)). \tag{4.2}$$

In particular, each of the summands on the right is smaller than or equal to $\mathrm{Var}(X)$. Note that it is possible that $\mathrm{Var}(X|Y=y) > \mathrm{Var}(Y)$ for some possible values for $y$ (as we saw above when $y = 2$) but it cannot be for all values for $y$.

The value of $\mathrm{Var}(\mathrm{E}(X|Y))$ is sometimes referred to as the *explained variance* (of $X$ by $Y$) as this is the part of $\mathrm{Var}(X)$ which is captured by $Y$. The rest of the variance, which is $\mathrm{E}(\mathrm{Var}(X|Y))$, is naturally referred to as the *unexplained variance*. The proof for (4.2) is deferred to the next section.

## 4.3 Covariance and correlation

The product $(x - \mathrm{E}(X))(y - \mathrm{E}(Y))$ is called the *covariate* at the pair $(x, y)$. Its sign says if the current $x$ and $y$ "go together" (when it is positive) or not (when it is negative). The magnitude of the products quantifies the relevant fact. When we look at all possible covariates we get a random variable $(X - \mathrm{E}(X))(Y - \mathrm{E}(Y))$ that has its own distribution.

The following table stays both the products and their corresponding probabilities for the example we defined above:

| X \ Y | 1 | 2 | 3 | |
|---|---|---|---|---|
| 1 | 1,0 | 0,1/6 | -1,1/6 | -1/6 |
| 2 | 0,1/6 | 0,0 | 0,1/6 | 0 |
| 3 | -1, 1/6 | 0, 1/6 | 1,0 | -1/6 |
| | -1/6 | 0 | -1/6 | -1/3 |

Note that in the margins we sum up the products between the covariates and their probabilities. Summing up the margins themselves leads us to $-1/3$ which is in fact $\mathrm{E}((X-\mathrm{E}(X))(Y-\mathrm{E}(Y))$. This value is called the *covariance* between $X$ and $Y$ and is denoted by $\mathrm{Cov}(X,Y)$. Note that the unit of measurement of the covariance is the product of the individual units. For example, if $X$ is in meters and $Y$ is in kg, then the covariance between $X$ and $Y$ is measured in meters times kilograms.

The correlation coefficient between $X$ and $Y$ is denoted by $\mathrm{Corr}(X,Y)$, and is defined as

$$\mathrm{Corr}(X,Y)=\frac{\mathrm{Cov}(X,Y)}{\mathrm{SD}(X)\mathrm{SD}(Y)}.$$

Note that this parameter is unit free and it tries to measure the *as-*

*sociation* between the two variables, an association which indeed got nothing to do with the units each of the variables is measured with. Also, note the symmetry: $\mathrm{Corr}(Y,X) = \mathrm{Corr}(X,Y)$.

A useful formula is

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) \pm 2\mathrm{Cov}(X,Y) + \mathrm{Var}(Y). \tag{4.3}$$

*Proof.*

$$\mathrm{Var}(X \pm Y) = \mathrm{E}(X+Y-(\mathrm{E}(X+Y)))^2 = \mathrm{E}(X-\mathrm{E}(X) \pm (Y-\mathrm{E}(Y)))^2 =$$

$$\mathrm{E}(X-\mathrm{E}(X))^2 \pm 2\mathrm{E}((X-\mathrm{E}(X))(Y-\mathrm{E}(Y))) + \mathrm{E}(Y-\mathrm{E}(Y))^2 =$$

$$\mathrm{Var}(X) \pm \mathrm{Cov}(X,Y) + \mathrm{Var}(Y).$$

□

Two random variables $X$ and $Y$ are said to be positively (negatively, respectively) correlated if $\mathrm{Cov}(X,Y) > 0$ ($\mathrm{Cov}(X,Y) < 0$, respectively). They are said to be *uncorrelated* if $\mathrm{Cov}(X,Y) = 0$. Note that if $X, Y$ are independent, then they are uncorrelated. The converse is not true: it is possible to construct example for uncorrelated pair of uncorrelated random variable which are not independent.

Back to the example:

$$\mathrm{Corr}(Y,Y) = \frac{-1/3}{\sqrt{2/3}\sqrt{2/3}} = -\frac{1}{2}.$$

In particular, $X$ and $Y$ are negatively correlated. This makes sense as we feel that the larger $Y$ is, the smaller should $X$ be.

It is well known that

$$-1 \le \mathrm{Corr}(X,Y) \le 1. \tag{4.4}$$

*Proof.* Note that if $ax^2 - 2bx + c \ge 0$ for any $x$, then $b^2 \le ac$. Equality might occur (and then it is at $x = b/a$) if and only if $b^2 = ac$. The following is a quadratic function in $\alpha$ which is never negative:

$$0 \le \mathrm{Var}(\alpha X - Y) = \mathrm{Var}(X)\alpha^2 - 2\mathrm{Cov}(X,Y)\alpha + \mathrm{Var}(Y).$$

Hence, $\mathrm{Cov}^2(X,Y) \le \mathrm{Var}(X)\mathrm{Var}(Y)$, which completes the proof. □

From the proof we learn that inequality (4.4) stands as equality if and only if for some $\alpha$, $\mathrm{Var}(\alpha X + Y) = 0$. This is possible if and only if $\alpha X + Y$ is a constant. In other words, $Y$ is a linear function of $X$ (or, equivalently, $X$ is a linear function of $Y$).

We next prove formula (4.2) which was stated above and its proof was deferred.

*Proof.*

$$\mathrm{Var}(X) = \mathrm{E}((X - \mathrm{E}(X))^2) = \mathrm{E}(\mathrm{E}((X - \mathrm{E}(X))^2)|Y)$$

$$= \mathrm{E}(\mathrm{E}((X - \mathrm{E}(X|Y) + \mathrm{E}(X|Y) - \mathrm{E}(X))^2)|Y)$$

$$= \mathrm{E}(\mathrm{E}((X-\mathrm{E}(X|Y))^2)|Y)+2\mathrm{E}((\mathrm{E}(X-\mathrm{E}(X|Y))(\mathrm{E}(X|Y)-\mathrm{E}(X)))|Y)+\mathrm{E}(\mathrm{E}(X|Y)-\mathrm{E}(X))^2)|Y)$$

$$\mathrm{E}(\mathrm{Var}(X|Y)) + 0 + \mathrm{Var}(\mathrm{E}(X|Y)).$$

We next argue why the second term equals zero. Note that in fact it is twice

$$\mathrm{Cov}(\mathrm{E}(X-\mathrm{E}(X|Y)), \mathrm{E}(X|Y)-\mathrm{E}(X)) = \mathrm{Cov}(\mathrm{E}(X-\mathrm{E}(X|Y)), \mathrm{E}(X|Y)).$$

Since

$$\mathrm{Cov}(\mathrm{E}(X|Y), \mathrm{E}(X|Y) - X) = \mathrm{Var}(\mathrm{E}(X|Y)) - \mathrm{Cov}(X, \mathrm{E}(X|Y)),$$

we need to show that the two terms in the right hand side are equal. Indeed,

$$\mathrm{Cov}(X, \mathrm{E}(X|Y)) = \mathrm{E}(X\mathrm{E}(X|Y)) - \mathrm{E}(X)\mathrm{E}(\mathrm{E}(X|Y)) =$$

$$\mathrm{E}(\mathrm{E}(X\mathrm{E}(X|Y)|Y)) - \mathrm{E}^2(X) = \mathrm{E}(\mathrm{E}(X|Y)\mathrm{E}(X|Y)) - \mathrm{E}^2(X)$$

(since $\mathrm{E}(X|Y)$ is constant when $Y$ is given)

$= \mathrm{E}((\mathrm{E}(X|Y))^2) - \mathrm{E}^2(X) = \mathrm{E}((\mathrm{E}(X|Y))^2) - \mathrm{E}^2(\mathrm{E}(X|Y)) = \mathrm{Var}(\mathrm{E}(X|Y))$,

as required.

□

Two random variables $X$ and $Y$ are said to be independent if one of the following equivalent properties hold: if for any pair of $x$ and $y$

1. $\mathrm{P}(X = x, Y = y) = \mathrm{P}(X = x)\mathrm{P}(Y = y)$
2. $\mathrm{P}(X = x|Y = y) = \mathrm{P}(X = x)$
3. $\mathrm{P}(Y = y|X = x) = \mathrm{P}(Y = y)$

Independence implies the following three properties (but the converse does not necessarily hold):

1. $\mathrm{E}(X|Y = y) = \mathrm{E}(X)$ for any $y$
2. $\mathrm{E}(Y|X = x) = \mathrm{E}(Y)$ for any $x$
3. $\mathrm{Cov}(X, Y) = 0$ and hence $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

# Chapter 5

# Families of discrete random variables

## 5.1 Moment generating functions

Suppose $X$ is a random variables that can receive only integer values (zero inclusive). Denote by $\mathrm{P}(X = i)$ by $p_i$, $i \geq 0$. Clearly, $\mathrm{E}(X) = \Sigma_{i=0} i p_i$, or in general $\mathrm{E}(X^k) = \Sigma_{i=0} i^k p_i$. Expected value can also define on functions of random variables. A function of interest $e^{sX}$ with $s$ is

a real number. Its expected value, which is of course a function of $s$, namely $\mathrm{E}(e^{sX})$, which is denoted by $M_X(s)$, is called the *moment generating function* (mgf) of $X$. The reason behind that if that

$$\frac{d^k}{s^k} M_X(s)|_{s=0}, \quad k \geq 0.$$

In other words, having the mgf of a random variable is equivalent to having all integral moments of it. As importantly, there is a one-on-one equivalents between these moments and the distribution (a claim we state without a proof). Hence, having the mgf in hand is, at least in principle, as having the distribution itself. Examples will be given shortly.

It is possible to see that

$$M_{aX+b}(s) = e^b M_X(as) \tag{5.1}$$

and that if $X$ and $Y$ are independent then

$$M_{X+Y}(s) = M_X(s) M_Y(s). \tag{5.2}$$

We next prove the later claim. Indeed,

$$M_{X+Y}(s) = \mathrm{E}(e^{s(X+Y)}) = \mathrm{E}(e^{sX} e^{sY}) = \mathrm{E}(e^{sX}) E(e^{sY}) = M_X(s) M_Y(s).$$

## 5.2 Bernoulli and binomial distributions

Suppose that $n$ is a natural number and that $p$ is some fraction in $[0,1]$. The random variable $X$ is said to be binomial with parameters $n, p$, if for each integer $k$, $0 \leq k \leq n$, its probability mass function is:

$$\mathrm{P}\left(X=k\right)=\left(\begin{array}{c} n \\ k \end{array}\right) p^{k}\left(1-p\right)^{n-k}.$$

These $n+1$ probabilities are positive and sum up to one. We denote a binomial random variable by $X \sim Bin\left(n,p\right)$.

We know that $\mathrm{E}(X) = np$, that $\mathrm{Var}(X) = np(1-p)$ and that $\mathrm{SD}(X) = \sqrt{np(1-p)}$. This random variable is used in order to count how many in a population of $n$ possess some property, where the individual probability is $p$ and where individuals are independent. Indeed, let $I_i = 1$ if individual $i$ possesses the property, $1 \leq i \leq n$. Otherwise, it equals zero. $I_i$ is referred to as an *indicator* random variable. Clearly, $I_i \sim Bin(1,p)$, $1 \leq i \leq n$, and this specific case of the binomial distribution is known as the Bernoulli distribution . By definition, $X = \Sigma_{i=1}^{n} I_i$. It is also easy to see that if $X \sim Bin(n,p)$ and $Y \sim Bin(m,p)$ and

they are independent, then $X + Y \sim Bin(n + m, p)$.

Suppose that $X \sim Bin(n, p)$. It holds that:

$$\frac{\mathrm{P}(X = k + 1)}{\mathrm{P}(X = k)} = \frac{\binom{n}{k+1}}{\binom{n}{k}} \cdot \frac{p^{k+1}(1-p)^{n-k-1}}{p^k(1-p)^{n-k}} = \frac{n-k}{k+1} \cdot \frac{p}{1-p}. \quad (5.3)$$

As long as this ratio is greater than one the probabilities are monotone increasing in $k$. The first $k$ under which this ratio is smaller than one, namely the $k$ which receives the highest probability, is known as the *mode*. It is to easy to see that it is the ceiling of $np+p-1$. Usually, there is a single peak but it is possible to "cook" examples where there are two peaks. For example, when $n = 9$ and $p = 0.5$, both 4 and 5 are the mode.

As for moment generating functions, note that

$$M_I(s) = e^{s0}(1-p) + e^{s1}p = (1-p) + pe^s.$$

Thus, by (5.2),

$$M_X(s) = ((1-p) + pe^s)^n. \quad (5.4)$$

Taking the first derivative

$$M'_X(s) = n((1-p) + pe^s)^{n-1} pe^s.$$

Hence,

$$\mathrm{E}(X) = M'_X(s)|_{s=0} = n(1-p+p)p = np.$$

## 5.3 Poisson distribution

An integer random variable $X$ is said to follow a Poisson distribution with parameter $\lambda$ is

$$\mathrm{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!},\ k \geq 0.$$

**Explanation for the Poisson distribution:**

We will show that the Poisson distribution is an approximation to the binomial distribution as $p \to 0$, $n \to \infty$ and $np \to \lambda$.
Denote $\lambda = np$, and note that if $X \sim Bin(n, p)$ then:

$$\frac{\mathrm{P}(X=k+1)}{\mathrm{P}(X=k)} = \frac{n-k}{k+1}\cdot\frac{p}{1-p} = \frac{np-kp}{(k+1)(1-p)} \overset{\substack{n\to\infty \\ p\to 0}}{\longrightarrow} \frac{\lambda}{k+1}$$

$$\Downarrow$$

So, in the limit we have

$$\mathrm{P}(X=k+1) \overset{\substack{n\to\infty \\ p\to 0}}{\longrightarrow} \frac{\lambda}{k+1}\cdot\mathrm{P}(X=k)$$

Therefore,

$$\mathrm{P}(X=k+1) = \frac{\lambda}{k+1}\cdot\frac{\lambda}{k}\cdot\mathrm{P}(X=k-1)$$

and so on. We can conclude by induction that:

$$\mathrm{P}(X=k) = \frac{\lambda^k}{k!}\cdot\mathrm{P}(X=0),\ \ k\geq 0$$

All that remains is to find the value of $\mathrm{P}(X=0)$. Indeed,

$$1 = \sum_{k=0}^{\infty}\mathrm{P}(X=k) = \sum_{k=0}^{\infty}\frac{\lambda^k}{k!}\cdot\mathrm{P}(X=0) = \mathrm{P}(X=0)\cdot\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \mathrm{P}(X=0)\cdot e^{\lambda}$$

Hence,

$$\mathrm{P}(X=0)=e^{-\lambda}$$

and then

$$\mathrm{P}(X=k)=\frac{\lambda^k}{k!}\cdot e^{-\lambda},\ k\geq 0.$$

It can also be concluded in the following way:

$$\mathrm{P}(X=0)=\binom{n}{0}p^0(1-p)^{n-0}=(1-p)^n=(1-\frac{\lambda}{n})^n\longrightarrow e^{-\lambda}$$

$$\mathrm{P}(X=1)=np(1-p)^{n-1}=\lambda(1-\tfrac{\lambda}{n})^{n-1}\to\lambda e^{-\lambda}$$

$$\mathrm{P}(X=2)=\binom{n}{2}p^2(1-p)^{n-2}=\tfrac{n(n-1)}{2}\cdot(\tfrac{\lambda}{n})^2(1-\tfrac{\lambda}{n})^{n-2}\to\tfrac{\lambda^2}{2}\cdot e^{-\lambda}$$

•

•

•

That is, for a large enough $n$, small enough $p$ and a medium $np$, the binomial distribution can be well approximated by the Poisson distribution.

It is possible to argue that $\mathrm{E}(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$ using the original definition for the Poisson probabilities. Alternatively, using the above proved limit results. Likewise, its mode equals the floor of $\lambda$.

**Claim:** Suppose $X$ and $Y$ are two independent random variables, both following a Poisson distributions with parameters $\lambda$ and $\mu$, respectively. Then, $X + Y \sim Pois(\lambda + \mu)$.

**Proof:**

$$\mathrm{P}(X+Y=k) = \sum_{i=-\infty}^{\infty} \mathrm{P}(X=i, Y=k-i) = \sum_{i=-\infty}^{\infty} \mathrm{P}(X=i)\mathrm{P}(Y=k-i)$$

$$= \sum_{i=0}^{k} \mathrm{P}(X=i)\mathrm{P}(Y=k-i) = \sum_{i=0}^{k} e^{-\lambda}\frac{\lambda^i}{i!}e^{-\mu}\frac{\mu^{k-i}}{(k-i)!}$$

$$e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^k}{k!}\sum_{i=0}^{k}\frac{k!}{i!(k-i)!}(\frac{\lambda}{\lambda+\mu})^i(\frac{\mu}{\lambda+\mu})^{k-i} = e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^k}{k!},$$

since the final summation is the sum of binomial probabilities with parameters $k$ and $\lambda/(\lambda+\mu)$. **END**

As for the msf, note that

$$M_X(s) = \sum_{i=0}^{\infty} e^{si}e^{-\lambda}\frac{\lambda^i}{i!} = e^{-\lambda}\sum_{i=0}^{\infty}\frac{(\lambda e^s)^i}{i!}$$

$$= e^{-\lambda}e^{\lambda e^s} = e^{-\lambda(1-e^s)}.$$

Thus,

$$M_{X+Y}(s) = e^{-\lambda(1-e^s)}e^{-\mu(1-e^s)} = e^{-(\lambda+\mu)(1-e^s)},$$

which is the mgf of a random variable whose distribution is Poisson with a parameter $\lambda + \mu$. This can be an alternative proof for the one given above. Finally, replace $\lambda$ in (5.4) and get

$$(1 - \frac{\lambda(1-e^s)}{n})^n.$$

Next, take the limit here when $n$ goes to infinity and get

$$e^{-\lambda(1-e^s)},$$

which, no surprise, the mfs for a Poisson random variable with parameter $\lambda$.

## 5.4 The geometric distribution

An integer value random variable $X$ is said to follow a geometric distribution with parameter $p$, $0 < p < 1$, if

$$\mathrm{P}(X = i) = p(1-p)^{i-1},\ i \geq 1.$$

The notation used is $X \sim Geom(p)$. $X$ counts the number of Bernoulli tries until and inclusive the first time of "success", namely that the value of corresponding indicator random variable equals 1. There is a version of it which replaces the "inclusive" with "exclusive". It is possible to see that $\mathrm{P}(X \geq i) = (1-p)^{i-1}$, $i \geq 1$. It is known that $\mathrm{E}(X) = 1/p$ and that $\mathrm{Var}(X) = (1-p)/p^2$.

**The hazard function.** In many cases a random variable states the longevity of an individual. It can the life span of a person (measured in years) or of a component. Denote $\mathrm{P}(X = i)$, by $p_i$, $i \geq 1$. Let then $q_i = \mathrm{P}(X \geq i) = \sum_{j=i}^{\infty} p_j$, $i \geq 1$, known as the *tail* probabilities. Finally, denote $\mathrm{P}(X = i | X \geq i) = p_i/q_i$ by $h_i$, $i \geq 1$. These probabilities are knows as the *hazard* probabilities. They give the probability that for one whose current age is $i$, namely has survived $i$ years, $i \geq 1$, this will be his final year. The complementary probability $1 - h_i$ is the survival probability for at least one more year. Clearly, $1 - h_i = q_{i+1}/q_i$, $i \geq 1$. Note, as it is easily to prove, that $p_i = \Pi_{j=1}^{i-1}(1-h_j)h_i$, $i \geq 1$. Indeed, in order for one to live exactly $i$ years, one needs to survive each of the first $i-1$ years and die at the year $i$, $i \geq 1$. In the area of actuary, the hazard is the single most important quantity as it determines the

yearly premium for life insurance.

It is possible to see that the hazard probabilities in the case of geometric distributions are constant and are equal to the parameter $p$. Indeed,

$$h_i = \frac{\mathrm{P}(X = i)}{\mathrm{P}(X \geq i)} = \frac{(1-p)^{i-1}p}{(1-p)^{i-1}} = p, \quad i \geq 1.$$

The converse is also true: If $X$ comes with a constant hazard, then it needs to follow a geometric distribution. Indeed, in this case denote by $h$ the constant hazard. Then,

$$\mathrm{P}(X = i) = \Pi_{j=1}^{i-1}(1-h_j)h_i = (1-h)^{i-1}h, \quad i \geq 1,$$

implying that $X \sim Geom(h)$. This fact show the one does not age or rejuvenate when one's life spam follows a geometric distribution. This is the *memoryless* property of this distribution. An alternative way to see this property is to note that

$$\mathrm{P}(X > k+l | X > l) = \frac{\mathrm{P}(X > k+l)}{\mathrm{P}(X > l)} = \frac{(1-p)^{k+l}}{(1-p)^{l}} = (1-p)^k = \mathrm{P}(X > k), \;\; k, l \geq 0.$$

In words, the probability that one will live an extra $k$ years is not a function of one's age $l$. Note that we have here

$$\mathrm{P}(X > k+l) = \mathrm{P}(X > l)\mathrm{P}(X > k), \quad k, l \geq 0,$$

which, at places, it is an alternative (but of course an equivalent) definition for memoryless.

As for the mgf:

$$M_X(s) = \sum_{i=1}^{\infty} e^{si} p(1-p)^{i-1} = pe^s \sum_{i=1}^{n} ((1-p)e^s)^{i-1} = \frac{pe^s}{1-(1-p)e^s}.$$

Note that

$$M'_X(s) = \frac{pe^s(1-(1-p)e^s) + pe^s(1-p)e^s}{(1-(1-p)e^s)^2}.$$

Plugging here $s = 0$ and we get that $\mathrm{E}(X) = 1/p$.